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Stump removal and fertilization of five *Phellinus weirii*-infested stands in Washington and Oregon affect mortality and growth of planted Douglas-fir 25 years after treatment

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Abstract

Phellinus weirii (Murr.) Gilb. causes laminated root rot (LRR), a major disease affecting growth and survival of *Pseudotsuga menziesii* (Mirb.) Franco (Douglas-fir) and other commercially important conifer species throughout the Pacific Northwest. Stump removal or nitrogen (N) fertilization, or both, have been suggested as management strategies for some root diseases but have not been tested previously for LRR in a replicated field study. Independent studies, using the same experimental design, were established in five Douglas-fir stands in Oregon and Washington. The stands were surveyed preharvest and postharvest (clearcut). Each *P. weirii*-infected entity (standing dead or down trees and stumps) was marked, and the location of its center was mapped. In each clearcut, a 2×4 set of factorial treatments of stump removal in combination with N fertilization was applied to 0.04 ha circular plots and replicated five to seven times. Treatments, applied in the fall, included stump removal with a bulldozer (either all stumps removed or the plot left undisturbed) and broadcast fertilization with ammonium nitrate (0, 336, 672, or 1345 kg N ha^{-1}). Douglas-fir seedlings were planted in the winter following treatment application. When the stand was considered established, each plot was thinned to a spacing of about 2.4 m inter-tree spacing and the trees tagged. Diameter at breast height, total height, and mortality of trees were recorded every 2–5 years. A total of 7827 tagged trees on 239 plots were observed for up to 27 growing seasons following treatment. Stumping caused a significant reduction in seedling mortality due to LRR at all five study areas. The mean LRR-caused mortality on stumped and nonstumped plots was 2.4 and 9.1%, respectively. Fertilization did not cause a significant change in LRR-caused mortality of seedlings at any of the study areas. Nitrogen fertilization before planting increased growth of Douglas-fir seedlings but only in some stands. Nitrogen fertilizer can have a positive residual effect on growth of Douglas-fir for at least 23 years. Effects on soil of heavy equipment used in stumping operations did not reduce subsequent Douglas-fir survival or growth. There was a mean reduction in wood volume of 24.8% in areas where LRR was observed compared to areas where it was not found.
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Keywords: Laminated root rot; Disease control; Nitrogen fertilization; Growth reduction; Stumping; Impacts; Epidemiology

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1. Introduction

Phellinus weirii (Murr.) Gilb. causes laminated root rot (LRR), a major disease affecting growth and survival of *Pseudotsuga menziesii* (Mirb.) Franco (Douglas-fir) and other commercially important conifer species throughout the Pacific Northwest and is responsible for large annual losses in stand productivity (Nelson et al., 1981; Thies, 1983; Bloomberg and Reynolds, 1985). It influences stand composition (Holah et al., 1993; Ingersoll et al., 1996) and dynamics (Cook, 1982; Holah et al., 1997) by directly killing susceptible trees or by predisposing them to windthrow, insect attack, and other secondary agents. *Phellinus weirii* spreads throughout a stand when uninfected roots contact roots of previously infected living or dead trees. *Phellinus weirii* can survive saprophytically in the roots of dead trees or stumps for several decades (Hansen, 1979); such persistence and spread allow the fungus to remain on the site beyond the life of an individual host and to infect trees of the replacement stand (Tkacz and Hansen, 1982). The biology, distribution, and impact of the disease, relative susceptibility of host species, and options for management are summarized in Thies and Sturrock (1995).

Removing stumps and roots from the soil (stumping) to reduce fungal inoculum has been suggested in the past as a management strategy for some root diseases. Stumping of infested sites has been proposed as an option to control annosus root disease in the southeastern United States (Kuhlman et al., 1976) and in the Pacific Northwest to reduce impacts of laminated root rot (Wallis, 1976; Thies, 1984; Hadfield, 1985; Morrison et al., 1988), armillaria root disease (Roth et al., 1977, 2000; Thies and Russell, 1984; Morrison et al., 1988), and annosus root disease (Hadfield et al., 1986).

The use of heavy equipment on forest soils, especially moist, medium- to fine-textured soils, can result in substantial soil compaction (Page-Dumroese, 1993). Growth of seedlings established on compacted soils is reported to be slower than for those on adjacent undisturbed soil (Youngberg, 1959; Froehlich, 1974; Froehlich et al., 1986; Smith and Wass, 1991); although on one site, growth was greatest in the most severe compaction treatment applied (Smith and Wass, 1984; Powers et al., 2004). Growth effects associated with compaction may last more than 30 years (Power, 1974; Wert and Thomas, 1981), and although natural recovery has occurred for some soils, the rates are slow (Froehlich et al., 1985; Powers et al., 2004). Besides causing negative physical changes in soil structure, compaction also may be responsible for negative biotic influences. Increased soil bulk density resulting from stumping and the associated machinery-induced vibrational compaction may reduce soil microbial activity (Dick et al., 1988; Shestak and Busse, 2005). In southern Oregon, soil compaction resulting from tractor logging is strongly correlated with black stain root disease, caused by *Leptographium wageneri* T.C. Harrington and F.W. Cobb, in Douglas-fir regeneration (Hansen et al., 1986).

Stumping does not remove all LRR inoculum from infested soil. Some residual infected roots are severed and remain. Broken ends and removed bark provide an opportunity for competing soil microorganisms to gain entrance and, under favorable circumstances, displace the pathogen. *Trichoderma* spp. are the primary fungi invading wood colonized by *P. weirii* (Nelson, 1964). The displacement process may be accelerated by applying high levels of nitrogen (N) fertilizer. In laboratory tests, N applied as either ammonium chloride or sodium nitrate dramatically reduced the viability of *P. weirii* in buried wood cubes (Nelson, 1970). Li et al. (1967) found that *P. weirii* cannot use nitrate N but antagonists such as *Streptomyces* spp. can. In field tests, applying urea reduced survival of *P. weirii* in wood (Nelson, 1975). Reduced *P. weirii* survival was later correlated with increased populations of *Trichoderma* species (Nelson, 1976).

If properly conducted, stumping and fertilization (either singly or in combination) may be useful in developing a strategy to reduce losses from LRR; however, timing of fertilization may be critical. Wallis and Reynolds (1974) found that fertilizing 26-year-old Douglas-fir, after *P. weirii* infection had already occurred, resulted in no impact on development of the fungus.

Accelerated growth of Douglas-fir seedlings for several years after application of N fertilizer is well documented. The addition of fertilizer pellets to planting holes before planting Douglas-fir gave mixed results on good sites (Austin and Strand, 1960; Arnott and Brett, 1973), and even on poor sites height growth increased for at least 5 years after application (Rothacher and Franklin, 1964). More recently, Douglas-fir seedlings

treated with polymer-coated fertilizer containing N had 11% greater stem diameter growth during the first 4 months after transplanting than did nontreated seedlings (Jacobs et al., 2005). Broadcast fertilization with ammonium nitrate several months before planting Douglas-fir resulted in increased growth of seedlings in two stands through their eighth and ninth growing seasons, respectively (Thies and Nelson, 1988; Thies et al., 1994). Broadcast fertilization of 225 kg N ha^{-1} is practiced in some Pacific coast stands of young to middle-age Douglas-fir (Bengtson, 1979; Tappeiner et al., 1986). Although duration of response to fertilization will differ owing to many factors, annual volume-growth response gradually approaches zero 10–15 years after treatment (Miller and Fight, 1979).

In 1976, we began a study to assess stumping or fertilizing, or both, with four dosages of ammonium nitrate as strategies for reducing the occurrence of LRR in replacement stands on infested sites. A secondary objective of the study was to assess the impact of these strategies on establishment and growth of seedlings. Although monitoring of the test stands to observe disease development is likely to continue, trends are clear enough now to draw some conclusions about the tested strategies. To allow a broad area of inference, the study was duplicated using the same design in five predominantly Douglas-fir stands in Oregon and Washington. Each stand is a complete, separate study. All stands received identical treatments but differed in the number of replicate plots, particulars about the parent stand, amount of LRR inoculum, and the year in which various activities occurred. Because the studies were identical in plan, they are presented together to allow the reader to consider region-wide similarities and draw general conclusions. In this paper, we report the effects of mechanically removing stumps from five sites, in combination with four application rates of ammonium nitrate, on the apparent occurrence and impact of LRR in the replacement stands and the growth of Douglas-fir regeneration 23–27 years after treatment.

2. Methods

2.1. Study areas

Each of the five study areas was established as an independent study with the same experimental design, but each was analyzed separately. Each study area comprised a single stand selected to be homogeneous for slope, soil, and general stand and understory vegetation characteristics. Blocking in each stand was based on relative amount of *P. weirii* inoculum on each plot and not on site characteristics. The number of replicates (Table 1), and thus the number of plots, was based on the size of each study area and availability of inoculum. Activities on each study area were conducted independent of the other sites. Thus, the activity schedules (Table 2) and the observation intervals (Table 3) were different. Because the individual studies had the same design and measured the same variables, some conclusions can be drawn that will have broad applicability. Each study area is identified by the name of a nearby town (see Fig. 1): *Apiary* – Apiary, OR; *Gates* – Gates, OR; *Hoodsport* – Hoodsport, WA; *La Grande* – La Grande, OR; and *Sweethome* – Sweethome, OR. The *La Grande* site is the only one east of the Cascade Range (dry side); the other four are west of the crest of the Cascade Range (wet side).

2.1.1. Apiary

The study area was a 4.5 ha clearcut on a 15% southwest-facing slope in the Oregon Coast Ranges (latitude 46°00′N; longitude 123°4′W); the mean elevation was 335 m. Mean annual precipitation measured near the site was 145 cm (United States Weather and Bureau, 1965). The soil in the study area was mapped as a Bacona silt loam with a surface layer of dark brown silt loam 15 cm thick. The upper layer of subsoil was 68 cm thick of reddish-brown silt loam and silty clay loam, and the lower subsoil was 68 cm

Table 1
Summary blocks, trees, and root systems by study area

Study area	Blocks[a]	Infected trees ha^{-1}		Examined root systems[b]
		Precut	Postcut	
Apiary	5	47.5	17.8	No count
Gates	7	32.6	26.7	3/71
Hoodsport	5	67.4	61.6	21/105
La Grande	7	126.0	19.8	No count
Sweethome	6	42.0	44.9	13/188

[a] Eight plots in each block.
[b] Stumps examined that were originally marked as without LRR and with complete root systems; no. of stumps with *P. weirii* on the root system (hidden infection)/no. root systems examined.

Table 2
Dates (month/year) of significant study activities

Study area	Stand level activity					Last observation	
	Cut	Treated	Planted	Tagging	Thinned	Ht/DBH	Mortality
Apiary	7/78	9/78	2/79	10/84	1/00	8/01	10/03
Gates	8/79	8/80	2/81	6/85	7/96	7/01	6/03
Hoodsport	8/76	10/76	3/77	9/81	10/99	8/01	10/03
La Grande	6/78	10/78	1/86	9/90	NA	6/02	6/02
Sweethome	7/79	8/80	1/81	7/85	NA	8/01	7/03

of yellowish-red silty clay loam. The effective rooting depth was 150 cm or more (Soil survey of Columbia County Oregon, November 1986). The site, site class II (McArdle et al., 1961), supported a 40-year-old naturally regenerated stand that was predominantly Douglas-fir with a lesser component of western hemlock (*Tsuga heterophylla* (Raf.) Sarg.), western redcedar (*Thuja plicata* Donn ex D. Don), chokecherry (*Prunus virginiana* L.), and red alder (*Alnus rubra* Bong.). The understory was primarily salal (*Gaultheria shallon* Pursh), vine maple (*Acer circinatum* Pursh) and swordfern (*Polystichum munitum* (Kaulf.) Presl).

2.1.2. Gates

The study area was a 6.1 ha clearcut on a 15% south-facing slope in the Cascade Range (latitude 44°46′N; longitude 122°21′W); the mean elevation was 603 m. Mean annual precipitation measured near the site was 200 cm (Silver Creek Falls, Oregon Station, Salem Fire Weather Office, National Weather Service). The soil in the study area was mapped as a Kinneyh cobbly loam (small percentage of McCully and Horeb soils); the surface was a very dark-brown cobbly loam about 25 cm thick; the subsoil, consisting of dark-brown cobbly clay loam in the upper part above a dark yellowish-brown cobbly clay loam in the lower part was about 75 cm thick, and the substratum was a dark yellowish-brown cobbly loam about 33 cm thick (Soil Survey of Marion County Area Oregon, Sept 1972). The site, site class II–III (McArdle et al., 1961), supported a 76-year-old naturally regenerated stand that was predominantly Douglas-fir mixed with western hemlock, western redcedar, and bigleaf maple (*Acer macrophyllum* Pursh). The understory was primarily swordfern, vine maple, salal, and Oregon grape (*Berberis nervosa* Pursh).

2.1.3. Hoodsport

The study area was a 3.8 ha clearcut on a 12% south-facing slope on the Olympic Peninsula (latitude 47°20′N; longitude 123°10′W); the mean elevation was 285 m. Mean annual precipitation measured near the site was 229 cm (Potlatch/Cushman Gauge Station, National Weather Service, Olympia, WA 98504). The soil in the study area was mapped as a Hoodsport gravelly sandy loam (state soil survey report for the south Puget Sound area, 1982, on file at the Forest Land Management division, Washington State Department of Natural Resources, Olympia, WA 98504). The Hoodsport soil series formed in glacial deposits of 50–75 cm of loose ablation till overlying

Table 3
Intervals (as seasons of growth) by study area for comparisons of mortality and growth

Time intervals for selected plot measurement activities	Number of seasons by study areas				
	Apiary	Gates	Hoodsport	La Grande	Sweethome
Height – planting to plot establishment	6	4	5	5	5
Mortality – planting to *prethin*	20	15	23	16	23
Height – planting to *prethin*	19	15	23	16	21
Posttreatment to start of *last increment*	19	15	21	NA	16
Length of last measured increment	4	6	4	NA	5
Treated to last observation	25	23	27	23	23

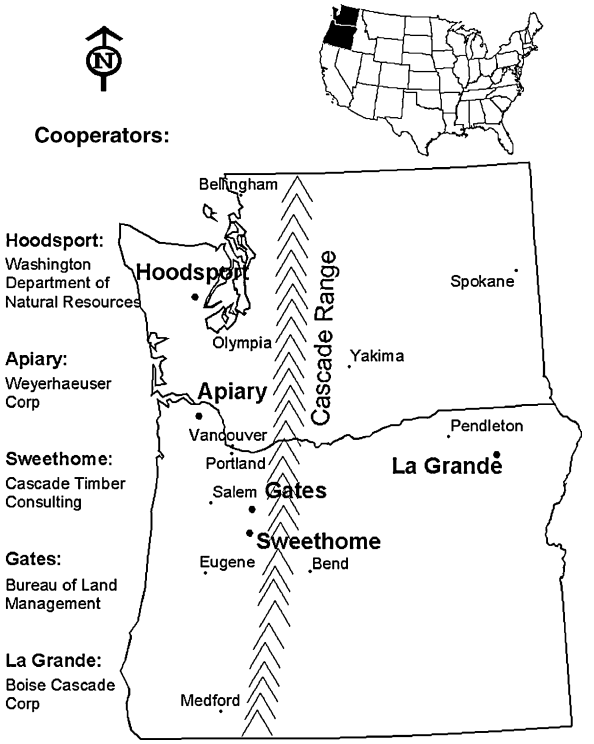


Fig. 1. Location of study areas.

very compact lodgement till. The site, site class II (McArdle et al., 1961), supported a 55-year-old naturally regenerated stand that was predominantly Douglas-fir (91% by volume) but also contained some western hemlock, western redcedar, and red alder in the overstory. The understory was primarily salal with some swordfern.

2.1.4. La Grande

The study area was a 5.5 ha clearcut on a 4% south-facing slope in the Blue Mountains, Oregon (latitude 45°32′N; longitude 118°27′W); the mean elevation was 1198 m. Mean annual precipitation measured near the site was 82 cm (including an average snowfall of 366 cm) (25-year average, Meacham Station, Oregon, National Weather Service). The soil in the study area was mapped as a Tolo silt loam (Soil survey of Umatilla County Area, Oregon, November 1988). This soil formed in volcanic ash over a buried soil, and the effective rooting depth was 150 cm. The surface layer was brown silt loam about 10 cm thick. The subsoil was very pale-brown silt loam about 45 cm thick. The next layer was a buried subsoil of light-yellowish-brown silt loam about 95 cm thick. The site, site class IV (McArdle et al., 1961), supported a mixed-age naturally regenerated stand with an overstory of grand fir (*Abies grandis* (Dougl.) Lindl.), Douglas-fir, western larch (*Larix occidentalis* Nutt.), ponderosa pine (*Pinus ponderosa* Dougl. ex Loud.), and lodgepole pine (*Pinus contorta* Dougl. ex Loud). The understory was primarily elk sedge (*Carex garberi* Fern) and twinflower (*Linnaea borealis* L.).

2.1.5. Sweethome

The study area was a 4.9 ha clearcut on a 16%, east-facing slope west of the crest of the Cascade Range, Oregon (latitude 44°21′N, longitude 122°39′W); mean elevation was 439 m. Mean annual precipitation measured near the site was 127 cm (25-year average, Sweet Home Ranger District, USDA Forest Service, Sweet Home, Oregon). The soil in the study area was mapped as the Honeygrove series (clayey, mixed, mesic, Typic Haplohumult), a silty clay loam derived from sedimentary and igneous colluvium (Langridge,

1987). The site, site class III (McArdle et al., 1961), supported a 60-year-old, naturally regenerated stand (following logging) that was predominantly Douglas-fir. Western hemlock, grand fir, western redcedar and red alder comprised the remainder of the overstory. The understory was primarily salal.

2.2. Plots

Each of the five study areas had the same plot layout and treatments, and each was subdivided into 30 m × 30 m units. Each unit was systematically surveyed and mapped on two separate occasions: first, before the then-existing stand was cut, to record infected standing dead or down trees or stumps (entities that could be moved during harvest); and, second, after harvest, to identify infected newly created stumps (Table 1). Each *P. weirii*-infected entity (standing or down tree, or stump) was marked and the location of its center mapped (Thies and Hoopes, 1979). Infected stumps or down trees were identified based on the presence of stain (incipient decay) or advanced decay typical of that caused by *P. weirii* or by presence of ectotrophic mycelia typical of *P. weirii* on roots near the root collar (Thies and Sturrock, 1995). As the fungus progresses through the roots and into the stem, newly invaded wood is stained. Trees with infected root tissue but without stain visible on the stump top (i.e. the fungus had not progressed that far) or ectotrophic mycelia visible at the root collar would not have been diagnosed and marked as infected. Thus, the survey represents a conservative estimate of the number of infected entities. Most infected entities would have been found, but an unknown number would have been missed. Identification of all infected stumps would have required total excavation and examination of all root systems, and this would have caused an unacceptable amount of disruption to the study.

An index of relative inoculum potential (INOC) was calculated for each infected entity. The index reflected the relative infected belowground stem and root biomass based on stump diameter 15 cm above the ground, the orientation of the stem, and an estimate of how long the associated tree had been dead based on retention of bark, twigs, and needles and stand entry records. At the beginning of this study, a model was not available to estimate belowground biomass of Douglas-fir given stump diameter for the size of trees in the parent stands. For this study, we used $BM = DSH^3$ as our estimator of relative biomass, where BM is the relative indicator of the belowground biomass associated with a particular stump that would be a target for uninfected roots to contact; and DSH is the diameter of the stem measured 15 cm above the ground. Our formula for inoculum potential was: $INOC = BM \times O \times Y$, where INOC is the relative biomass of potential inoculum, O is the orientation (factors: standing = 1.00; leaning, roots partially out of the ground = 0.66; down, tree on the ground = 0.20); and Y is the years dead (factors: live = 0.60, dead 0–5 years = 0.95, dead 6–10 years = 0.90, dead more than 10 years = 0.50, stump left from an earlier thinning = 0.10). Factor values were based on the senior author's experience with excavating hundreds of Douglas-fir stumps and thousands of roots.

On a map depicting the locations of infected trees or stumps within each study area, points were marked as centers for 0.04 ha circular, nonoverlapping treatment plots: eight plots were positioned over areas without known inoculum, additional plots were positioned to cover concentrations of known *P. weirii* inoculum. Centered within each 0.04 ha circular treatment plot was a 0.02 ha circular seedling-measurement (data) plot. This resulted in a 3 m-wide treated buffer around each 0.02 ha measurement plot. Each measurement plot was rated for inoculum potential by totaling the INOC for stumps within the plot and adding 25% of the INOC for stumps within the buffer. We assumed that an average of 25% of the roots of infected trees in the buffer would extend into the measurement plot thus increasing available inoculum. Plots were stratified based on total INOC into blocks of eight plots each. The number of plots was limited by the number and distribution of infected entities. The number of blocks depended on the number of plots in a particular study area (Table 1). The range of INOC ratings within a block was reduced to a minimum by shifting some plots on the map, thus changing their INOC rating. Based on the map, plot centers were temporarily established on the ground, and the location of infected stumps on each plot was verified visually. After stump removal, each plot center was reestablished and marked by a steel rod.

2.3. Treatments

Using a factorial design, we applied treatments in late summer that consisted of stumping in combination with N fertilization. Two stumping treatments were used: S_0, no stumps removed, and S_1, all stumps removed. Stumping was done to remove as much of the stump and root system as possible through operational techniques; however, no secondary effort was made to remove severed roots from the soil. A crawler tractor (32,000–40,000 kg) equipped with either a solid or brush blade on the front and usually having a splitting wedge on the back was used. Stumps larger than 45 cm in diameter were first split at least once by backing into them with the wedge or by pushing with a corner of the blade. Stump pieces were then pushed out with the blade. Whenever possible, stumps were left upended where they came to rest. For all sites stumping was done when the soil surface was dry. Because of this dry condition, little soil clung to roots. By visual assessment, large-scale lateral soil movement seemed minimal; however, substantial mixing of topsoil and subsoil seemed to occur, and nearly all vegetation on the stumped plots was destroyed.

All excavated stumps and root systems from three of the five study areas were examined more closely. Stumps originally marked as not infected by *P. weirii,* based on stump surface examination and for which complete root systems could be found, were assessed. Roots from these stumps were cut every 30 cm and examined for stain or decay typical of *P. weirii*. Stumps with positive evidence of infection by *P. weirii* were termed hidden infection (Table 1). In addition, 40 hollow stumps with LRR were examined.

Ammonium nitrate, as small prills, was applied at four levels: F_0, no fertilizer applied; F_1, 336 kg N ha^{-1}; F_2, 672 kg N ha^{-1}; and F_3, 1345 kg N ha^{-1} (0, 300, 600, and 1200 lb N $acre^{-1}$, respectively). The fertilizer was broadcast by a person using a cyclone-type seed spreader. Half was applied by walking parallel transects across the plot. The remainder was spread similarly by crossing the plot along transects perpendicular to the direction of the first lines. This application technique ensured even application. The eight possible treatment combinations of stumping and fertilizing (S_0F_0, S_0F_1, S_0F_2, S_0F_3, S_1F_0, S_1F_1, S_1F_2, S_1F_3) were randomly assigned within each block of eight plots.

2.4. Seedling establishment

Douglas-fir seedlings (2-1 bare-root) from a local seed source were planted on each study unit in the winter immediately following treatment application. To compensate for losses to animal browsing or damage during planting, plots were examined the first fall after planting and interplanted to achieve a minimum density of 2200 seedlings ha^{-1}. The *La Grande* study site was an exception. This site suffered repeated extreme browse damage, and seedlings were interplanted four times between 1978 and 1986. The site was fenced in 1984 to help achieve an adequate density of live seedlings. From planting until the seedlings were considered established, the plots were visited at intervals of 2 years or less, seedling mortality from LRR was determined and recorded (but other causes of mortality were not recorded), and volunteer trees and competing brush were removed.

A study area was declared established when a majority of the seedlings were vigorous and considered large enough to be unlikely to suffer significant browse damage; this usually took five growing seasons. During the summer the stand was declared established, excess seedlings were pulled by hand to leave a final density of 34 Douglas-fir seedlings on each 0.02 ha measurement plot (1700 seedlings ha^{-1}). Retained seedlings were selected to achieve even spacing (approximately 2.4 m intertree spacing) over the entire plot (first priority) and to leave the most vigorous seedlings (second priority). The same selection criteria were applied to all plots at all study areas. The roots of removed seedlings were examined for evidence of LRR. Each retained seedling was marked with a numbered aluminum tag, and its location on the plot was mapped. These plot establishment activities and the date they took place will be referred to here as *tagging*. With the exception of the *La Grande* site, all measurement plots on all study areas had 34 seedlings at *tagging*. Most of the *La Grande* plots had less than 34 seedlings.

Subsequent management activities at each study area, such as brush control or precommercial thinning, were prescribed and conducted by the local land

manager and were applied uniformly to all parts of the particular study area without regard to plots. Three of the study areas were precommercially thinned (Table 2) to a final density as follows: Apiary, 1013 trees ha^{-1}; Gates, 687 trees ha^{-1}; and Hoodsport, 603 trees ha^{-1}. Determinations of density and selection of trees to be retained on each plot in a study area were made given the same criteria regardless of study treatment.

Because of change in drainage associated with road construction at the *Hoodsport* site, one plot, treatment S_0F_1 replicate five (no infected stumps), was underwater most of the year. Because of excess water, most seedlings were dead at *tagging* and the plot was abandoned. A missing-plot formula was used to calculate a substitute value for this plot for analyses.

2.5. Data collection and analysis

At *tagging*, height and diameter at breast height (DBH, 1.4 m above the ground) of each plot tree were recorded. If a seedling showed signs of severe animal damage it was not included in the calculation for the plot mean height. This happened to less than 3% of the trees at *tagging*, and these trees were not clustered in any one treatment or block. After *tagging*, plots were examined at intervals of 2–5 years for the next 12 years (La Grande) to 20 years (Hoodsport), tree mortality from LRR was recorded, and volunteer trees and competing brush were removed. Time interval between plot activities was variable, differed for each study area, and was dependent on available resources; however, mortality was evaluated on up to 10 occasions per study area, and height and DBH were evaluated on up to six occasions per study area. The interval between evaluations was kept relatively short to capture as much information as possible in the event that trees on all or part of a study area were unexpectedly destroyed. The DBH and height of each live tree were used to estimate tree volume (Bruce and DeMars, 1974).

Three sets of data collected at comparable stages of stand development for each study area were examined for this paper: (1) *tagging* – tree heights collected at time of *tagging*; (2) *prethin* – the last data set collected from the remaining original 34 tagged seedlings on each plot (data used for the three thinned study areas [*Apiary*, *Gates* and *Hoodsport*] were taken before thinning, data for the two as yet unthinned study areas [*La Grande* and *Sweethome*] were the last measurements taken Table 2) and; (3) *last increment* – last 4 or 6 years of growth of each tree, calculated for each study area (except La Grande) from the last two data sets.

Four variables, as plot means, were analyzed to evaluate the effect of the stumping and fertilization treatments on the trees at each study area: (1) tree height at *tagging*; (2) mortality – the proportion of the 34 tagged seedlings on each plot that died from LRR using *prethin* data; (3) Mean height of surviving trees on each plot using *prethin* data; and (4) final increment of plot volume. Tree volume was determined for individual Douglas-fir (Bruce and DeMars, 1974) by using *last increment* data. The difference between the volume at the beginning and end of the increment became the final increment of volume growth for that tree. Tree data were aggregated for final plot volume and volume of final increment of plot growth (m^3/ha).

Analysis of variance (ANOVA) was used to test plot means for seedling height at *tagging*, proportion of mortality due to LRR, tree height at *prethin*, and final increment of tree volume growth. All proportion data were transformed using an arcsine square root transformation before ANOVA evaluation. When the fertilization treatment caused significant differences, comparison of the means for the four fertilization levels was conducted by using Tukey's test for multiple comparisons (Ramsey and Schafer, 1997). Interactions between fertilization and stumping were tested by ANOVA. All statistical analyses were conducted using S-PLUS 6.1 (Insightful, 2002). Differences were considered to be significant when $p \leq 0.05$.

Usefulness of either INOC or a count of infected entities from the parent stand as a predictor of seedling mortality were evaluated by using only data from plots where the treatments did not impact seedling mortality from LRR. Analysis was by multiple linear regression of the arcsine square root transformation of the proportion of LRR mortality.

Data from 119 nonstumped plots (3888 tagged trees) and 120 stumped plots (3939 tagged trees) were used to evaluate the impact of LRR and the stumping treatment on plot volume. We compared four groups of plots: nonstumped without LRR ($n = 20$),

Table 4
Significance results for ANOVA of stumping treatments

Variable tested	*p*-Value[a] by study areas				
	Apiary	Gates	Hoodsport	La Grande	Sweethome
Tree height at *tagging* (m)	–	–	<0.0001	–	–
Mortality due to LRR (%)	0.0008	0.0027	0.0069	0.0007	0.0021
Tree height at *prethin* (m)	–	0.0141	0.0218	–	–
Final plot growth increment (m^3/ha)	0.0374	###	–	–	–

###, $0.05 < p$-value < 0.10; – = p-value > 0.10.
[a] Cells contain p-value where significant ($p < 0.05$) differences were found.

nonstumped with LRR ($n = 99$), stumped without LRR ($n = 19$), and stumped with LRR ($n = 101$). A plot was classified as with LRR if the disease was found either preharvest or postharvest (parent stand) or if there was any LRR caused mortality of tagged trees on the plot (replacement stand). Volumes used were those calculated from *prethin* data. The group means analysis was done with ANOVA (Ramsey and Schafer, 1997).

3. Results

A total of 7827 Douglas-fir seedlings were tagged, measured, and observed for up to 27 growing seasons. Of the 871 tagged trees that died, 400 died from LRR. Plot means are summarized by both study area and treatment level for those treatments showing significant differences (stumping Tables 4 and 5, fertilization Tables 6 and 7).

Stumping caused a significant increase in the mean height of seedlings measured at *tagging* at *Hoodsport* but no significant change at the other four study areas (Table 4). Fertilization did not cause a significant change in mean height of seedlings measured at *tagging* at any of the study areas (Table 6).

Stumping caused a significant reduction in seedling mortality due to LRR at all five study areas (Tables 4 and 5). The mean LRR-caused mortality on stumped and nonstumped plots was 2.4 and 9.1%, respectively. Fertilization did not cause a significant change in LRR-caused mortality of seedlings at any of the study areas (Table 6).

Stumping caused a significant increase in the mean height of seedlings measured *prethin* at *Hoodsport* and *Gates* but no significant change at the other three study

Table 5
Plot means and standard errors

Treatment	Plot means and standard errors by study area				
	Apiary	Gates	Hoodsport	La Grande	Sweethome
Tree height at *tagging* (m)					
S_0	–	–	1.36 (0.05)	–	–
S_1	–	–	1.72 (0.05)	–	–
Mortality due to LRR (%)					
S_0	7.21 (1.66)	3.47 (1.22)	22.29 (4.69)	4.16 (1.03)	11.40 (2.53)
S_1	1.91 (1.09)	0.32 (0.18)	6.80 (1.58)	0.65 (0.32)	2.08 (0.93)
Tree height at *prethin* (m)					
S_0	–	9.79 (0.15)	13.73 (0.48)	–	–
S_1	–	10.32 (0.16)	14.98 (0.37)	–	–
Final plot growth increment (m^3/ha)					
S_0	6.05 (0.26)	–	–	–	–
S_1	5.34 (0.27)	–	–	–	–

S_0, Nonstumped plots; S_1, stumped plots; – differences between treatments are not significant; the value in parenthesis where differences between stumping treatments were significant.

Table 6
Significance results for ANOVA of fertilization treatments

Variable	*p*-Value[a] by study areas				
	Apiary	Gates	Hoodsport	La Grande	Sweethome
Tree height at *tagging* (m)	###	–	–	–	–
Mortality due to LRR (%)	–	–	–	–	–
Tree height at *prethin* (m)	–	–	–	–	0.0198
Final plot growth increment (m^3/ha)	0.0053	–	###	N/A	0.0011

###, $0.05 < p$-value < 0.10; –, p-value > 0.10.
[a] Cells contain p-value where significant ($p < 0.05$) differences were found.

areas (Table 4). Fertilization caused a significant change in the mean *prethin* height of seedlings at *Sweethome* but not at any of the other study areas (Table 6). At *Sweethome,* the F_3 treatment resulted in significantly greater height than the F_0 treatment. The other fertilizer treatments at *Sweethome* did not cause any significant change (Table 7).

Stumping caused a significant decrease in the mean final increment of plot volume at *Apiary* but did not cause a significant change at any of the other study areas (Table 4). Fertilization caused a significant increase in mean final increment of plot volume at *Apiary* and *Sweethome* but not at any other study area (Table 6). At *Apiary*, the highest rate of fertilizer treatment (F_3) caused a significantly greater mean final increment of plot volume than the no fertilizer treatment (F_0) but the other fertilizer treatments did not cause any significant change. At *Sweethome* the F_3 and F_2 treatments caused a significantly greater mean final increment of plot volume than the F_0 treatment, but the other fertilizer treatment did not cause any significant change (Table 7). Final growth increment was not determined for the La Grande study area.

Differences among replicates (blocked on INOC) were not significant for any of the variables tested at any study area. We did not detect significant interactions between stumping and fertilization treatments.

Because the fertilizer treatments did not cause significant differences in the proportion of mortality observed at any study area, all nonstumped (S_0) plots ($n = 119$) were used to test both the ability of INOC or a simple count of infected stumps in the parent stand to predict the proportion of *P. weirii*-killed seedlings in

Table 7
Plot means and standard errors

Treatment	Plot means and standard errors by study area				
	Apiary	Gates	Hoodsport	La Grande	Sweethome
Tree height at *tagging* (m)	–	–	–	–	–
Mortality due to LRR (%)	–	–	–	–	–
Tree height at *prethin* (m)					
F_0	–	–	–	–	13.00 (0.41) a
F_1	–	–	–	–	13.87 (0.34) ab
F_2	–	–	–	–	14.42 (0.36) ab
F_3	–	–	–	–	14.84 (0.51) b
Final plot growth increment (m^3/ha)					
F_0	4.94 (0.29) a	–	–	N/A	2.12 (0.18) a
F_1	5.43 (0.27) a	–	–	N/A	2.57 (0.23) ab
F_2	5.93 (0.33) ab	–	–	N/A	3.16 (0.19) b
F_3	6.92 (0.44) b	–	–	N/A	3.17 (0.23) b

(a) and (b) means within columns and within the same variable followed by the same letter are not significantly different ($p > 0.05$);–, fertilizer treatment was found to be not significantly different from no treatment. The values in the parenthesis where differences between fertilization treatments were significant ($p < 0.05$). F_0, no fertilizer applied; F_1, 336 kg N ha^{-1}; F_2, 672 kg N ha^{-1}; and F_3, 1345 kg N ha^{-1} (0, 300, 600, and 1200 lb N $acre^{-1}$).

the replacement stand. Although both models had slopes significantly different from zero ($p < 0.0001$), there is too much unexplained variation in the models for either INOC or a count of infected stumps to be useful in predicting proportion of mortality in the replacement stand ($R^2 = 0.3057$ and 0.3876, respectively). Of the 119 nonstumped plots, six of the 25 plots with no LRR detected in the parent stand had LRR-killed trees in the replacement stand, and 29 of the 94 plots with LRR detected in the parent stand did not have LRR-killed trees in the replacement stand. Of the 120 stumped plots, three of the 22 plots with no LRR detected in the parent stand had LRR-killed trees in the replacement stand, and 65 of the 98 plots with LRR detected in the parent stand did not have LRR-killed trees in the replacement stand.

Plots with LRR contained less volume of Douglas-fir than plots without LRR. After accounting for study area and fertilizer level, there is convincing evidence that the presence of detectable LRR in a plot accounted for a reduction in the mean prethin volume (two-sided $p < 0.0001$). The estimated reduction is 34.6 m^3/ha (se = 8.15 m^3/ha). This equals a prethin stand volume reduction due to the presence of LRR of 24.8%. There also is convincing evidence that stumping of LRR-detected plots increased mean prethin plot volume (two sided $p < 0.0001$). The estimated volume increase was 23.8 m^3/ha (se = 4.57 m^3/ha). This is equivalent to a prethin volume increase due to stumping of 25.4% on LRR infested plots. There was no evidence of a volume difference between nonstumped without LRR plots and stumped with LRR plots ($p = 0.19$).

At the time *Hoodsport* was harvested (1976), black stain root disease was found in a few live Douglas-fir. Since then we have observed mortality from this disease in the planted Douglas-fir at all ages. The trend is for an increasing rate of mortality as the stand ages. We have not detected evidence that any of the treatments in our study have altered the mortality rate from black stain root disease.

The excavated hollow stumps had an opening delineated by one or two growth rings and the inside of each hollow had traces of advanced decay and setal hyphae typical of *P. weirii*. Examined stumps were found to have few if any infected roots. In many instances the hollow was half the diameter of the stump and in some cases extended more than 1.5 m up the stem and down into the root collar.

4. Discussion

This paper reports on results from five independent, complete studies that examined the effects of stump removal and fertilization on mortality and growth of planted Douglas-fir. Even though each study was installed independently without regard to the others, all studies shared a common experimental design. Each study site can be evaluated on its own merit, and the results of testing four variables by ANOVA for each study area are provided (Tables 4 and 6). By reporting results from the five sites together, we expose the limitations of examining LRR and tree responses to treatments in only one stand. We also expand our area of inference and are able to address some questions with a larger data set. The results provided in Tables 4 and 6 illustrate the value of a multistand approach.

Stumping consistently reduced the reappearance of LRR in the replacement stands. Tree mortality, due to LRR, was significantly less during the observation period from *tagging* (stand age 5 or 6 years) through *prethin* (11–18 years later) on stumped (2.4%) than on nonstumped (9.1%) plots in all study areas. Quantitative data were not collected on seedling mortality prior to *tagging*; however, many of the dead seedlings examined, even at the end of the first growing season, and live excess seedlings removed at *tagging* had LRR. Most of these infected seedlings were from nonstumped plots.

If land managers conduct stumping with a bulldozer to manage LRR, as was done in the studies reported here, they should expect that some LRR will appear in the replacement stands. Of the 120 stumped plots, across all study sites, 22 were rated as LRR free in the parent stand, yet LRR-caused mortality occurred on three of these 22 plots. Of the 98 stumped plots determined to have LRR inoculum in the parent stand, the disease reappeared on 33 (about one-third) of all treated plots. Stumping usually breaks off some infected roots, and some remain in the soil to serve as inoculum in the replacement stand. Soil displaced during stumping also may obscure some stumps and cause them to be missed by the equipment

operator and not removed; some of these missed stumps likely will be infected. A more rigorous removal of *P. weirii* inoculum than reported here involves stumping followed by root raking to remove broken root pieces. Even with this treatment some LRR reappears in the replacement stand (Morrison et al., 1988). In our study, LRR-killed trees occurred on one-third of the stumped plots in the replacement stand. We speculate that with normal spread of this disease, this distribution of infected trees, over time may diminish the anticipated disease-reduction benefits of stumping as reported here.

Since the initiation of our study, other methods of stump removal have been proposed and tested that seem to leave fewer residual infected roots in the soil. Use of wide-track excavators to push-fall trees or remove stumps after conventional logging has many advantages over the more common bulldozer with brush blade used in this study (Morrison et al., 1992; Sturrock et al., 1994; Thies and Sturrock, 1995). A thorough evaluation of the efficacy of newer methods to mechanically remove *P. weirii* inoculum will require additional time for field studies to mature.

It was anticipated that stumping would result in soil compaction that could cause reduced tree growth. Our plots probably received the maximum amount of compaction likely to occur during stump removal with a bulldozer given similar sites and moisture conditions. Our stumped areas were relatively small (0.04 ha) compared with an area to be stumped operationally, and because they were research plots, many stumps were pushed completely off plots to assure space for the desired planting density. Also, the bulldozer had to remain within the boundary of the treatment plot to avoid impacting nearby nonstumped plots. As a result, the bulldozer repeatedly travelled over nearly every square meter of each plot. Other investigators have observed a negative effect on seedling growth correlated with the operation of heavy equipment on some forest soils and have generally attributed the response to soil compaction. The effects of compaction and methods proposed to ameliorate those effects are reviewed elsewhere (Lull, 1959; Froehlich, 1985; Froehlich and McNabb, 1984).

In our study, stumping did not have the anticipated negative impact on height growth of Douglas-fir trees in the replacement stand. In two areas, Gates and Hoodsport, stumping increased height growth as measured just before thinning. The only evidence of a negative impact on tree growth from stumping was a decrease in the final increment of volume growth at Apiary; however, the decrease was small, the p-value was marginally significant, and analysis of all earlier measures of tree-height growth at Apiary indicated no significant difference between stumped and non-stumped plots. We believe that the decrease in final increment volume at Apiary was not biologically significant.

Morrison et al. (1988) measured the growth of Douglas-fir seedlings 20 years after treatment on a 1.3 ha tract in British Columbia that had been stumped and root-raked to remove LRR inoculum. They reported better growth on the stumped tract than on an adjacent nonstumped tract. Although we do not have data on pore-size in the soils for our five study areas, soil compaction may improve water-holding capacity by altering the pore-size distribution, thereby minimizing negative effects on seedling growth (Gomez et al., 2002). At this time, we lack information to explain increased tree height growth with the compaction we observed.

Although all study areas were anticipated to have increases in soil bulk density owing to compaction, it seems that either compaction was not as great as expected or recovery was faster. Soil compaction was not evaluated at the time the treatments were applied, but it has been evaluated since. In a separate study, soil compaction on these same sites was tested in 2003, and even though the mean bulk density on the stumped plots was 3% higher than on the nonstumped plots, the differences were not statistically significant (D. Chambreau, University of Washington, 2005, personal communication). Repeated bulk density measurements from *Sweethome* showed recovery from increases caused by stumping. Data taken in 1991 (Thies et al., 1994) show bulk density on stumped plots to be 7% higher than on nonstumped plots ($p = 0.016$). Repeat measurements on the same plots show that the stumped and nonstumped plots are now similar in soil bulk density, 0.97 and 0.96 g cm^{-3}, respectively, and are similar to the surrounding undisturbed forest land (D. Chambreau, University of Washington, 2005, personal communication). These recovery rates for surface soils are faster than those found in other studies that examined compaction

on skid trails (Wert and Thomas, 1981; Froehlich et al., 1985).

Fertilizer, as applied, did not have a detectable effect on development of LRR on planted seedlings. These results were consistent for all five-study areas. Our highest fertilizer application rate was chosen to test the disease-reduction potential of the maximum rate we felt could be reasonably (physically and biologically) applied to a site. We concluded that N fertilization, as applied in this study, is not an effective management strategy for LRR.

Preplant fertilization with ammonium nitrate did not cause a positive growth response in planted Douglas-fir in the short term, but it did in the longer term. Seedling height, measured at *tagging*, was not significantly different on fertilized plots than on nonfertilized plots at any of the five study areas. By *prethin* at *Sweethome,* the mean tree height was significantly greater on plots fertilized at the highest rates as compared to nonfertilized plots (Tables 6 and 7). At *Apiary* and *Sweethome* there was evidence of a significant increase in the final increment of volume growth (29 and 33%, respectively) on the plots fertilized at the highest rates as compared to the nonfertilized plots. This was the growth during the final 4–6 years of our observations starting 16 or 19 years after treatment. We interpret this to mean that the fertilizer is continuing to have an effect on tree growth on two of the stands and that the greater mean tree height at *Sweethome* is not just a result of early growth immediately following fertilization. We concluded that there is a residual effect of the fertilizer application that is measurable as increased growth rates starting 19 years after treatment.

We are unaware of other studies that have tested the effects of broadcast fertilization of sites before planting with Douglas-fir or of any that report effects on trees similar in age to those in our study. Addition of a fertilizer pellet to planting holes before planting Douglas-fir seedlings gave mixed results on good sites (Austin and Strand, 1960; Arnott and Brett, 1973) but it increased height growth for at least 5 years on unfavorable sites (Rothacher and Franklin, 1964). Application of 225 kg N ha^{-1} is a common practice on private lands in some Pacific Coast stands of young to middle-aged Douglas-fir (Bengtson, 1979; Tappeiner et al., 1986). Initial applications may be conducted in stands aged 15–20 years old and may be repeated every 5–10 years. Although the duration of responses will differ depending on many factors, growth responses gradually approach zero within 10–15 years after fertilization (Miller and Fight, 1979). Our lowest fertilizer rate was within the biologically and economically optimum fertilizer dosage recommended for use in established stands of Douglas-fir (Miller and Fight, 1979). This low rate did not significantly increase tree volume over the nonfertilized plots for either height at *prethin* (*Sweethome*) or last increment of volume (*Apiary* and *Sweethome*). The highest two rates that we applied did demonstrate an effect on growth for 23 years however, well beyond the 10- to 15-year duration of effect reported by Miller and Fight (1979).

Data from our five study sites provided a unique opportunity to examine the impact of LRR on volume growth of Douglas-fir. Previously reported estimates of reduced growth of Douglas-fir caused by LRR were of three types: (1) intensive examination of a limited number of individual trees (Bloomberg and Wallis, 1979; Thies, 1983); (2) region-wide estimates of loss based on personal observation and a limited amount of survey data (Childs and Shea, 1967; Nelson et al., 1981); or (3) estimates of the occurrence of LRR on a particular type of ownership (Gedney, 1981). Our study provided an opportunity to compare growth records of 3888 trees on the 119 nonstumped plots with and without LRR within the same stands. Plot locations were selected without knowledge of potential tree growth. We classified plots based on the presence of LRR both in the preceding stand and the current stand. We found that the presence of LRR reduced stand mean volumes 24.8% based on undisturbed plots across five stands in the Pacific Northwest. A similar classification and comparison of volumes on stumped plots found no difference between plots with LRR and those without, further evidence that stumping reduces losses from LRR. A comparison of volumes of plots with LRR found that stumping increased the volume on plots with LRR by 25.4%.

Although this volume reduction is based on data from young stands, we see no reason why this percentage of loss should be expected to diminish as stands age. Rather, we expect that reduced volume growth would continue and would likely decrease further with additional standing mortality and wind-

throw as infected roots die and lose their structural integrity. Mortality alone (9.1% on nonstumped plots) cannot account for the reduced stand volume. If the estimated LRR-caused reduction in growth for all sites in this study was about 15.7% (24.8% volume loss total minus 9.1% due to mortality), it would be of the same general magnitude found in earlier studies (Bloomberg and Wallis, 1979; Thies, 1983). We conclude that managers can use 24.8% as a reasonable estimate of the reduced volume in those portions of young Pacific Northwest Douglas-fir stands where LRR has been observed. The rate of LRR-caused mortality and reduced growth of other host species may be different and needs further research.

The failure of both INOC (an indicator of the amount of inoculum) and a simple count of known infected stumps (an estimator of distribution or density) as predictors of mortality in the replacement stand demonstrate the difficulty of surveying for LRR in particular and root diseases in general. Our approach was to check carefully any dead trees and to mark any stumps having stain typical of LRR, yet there still were hidden infections that apparently provided viable inoculum that we could not account for. When we examined excavated stumps we found 37 of 364 (about 10%, Table 1) of the stumps that we had marked as not infected actually had LRR. We examined only excavated stumps with complete root systems. Stumps with decayed roots that easily separated from the stump during removal were not examined. Thus, the estimated 10% misclassification rate is itself low. Although intuitively we expect that more infected stumps in a stand will result in more infected trees in the replacement stand, at this time we find a low correlation between the inoculum observed or inferred and the number of infected trees that can be expected in the replacement stand. Were this low correlation supported by additional field studies, it would suggest a need to re-evaluate existing root disease models.

All infected stumps may not act as inoculum. In this study, we observed 40 hollow stumps that were infected but did not have signs of currently infected roots and thus were unlikely to act as inoculum. In an earlier study, Thies (1982) reported that of 111 live-infected plot trees, 13 were found to have a hollow stump. Few roots on the hollow stumps were infected, and it was not possible to isolate *P. weirii* from wood collected more than a few millimeters beyond the limit of the hollow. This limitation of the growth of the pathogen in the host occurs on less than 15% of the observed infected trees and only superficially resembles compartmentalization as described for some species in the eastern United States (Shigo, 1979a, b; Shigo and Tippett, 1981). Further, it was found that trees with a hollow in the base exhibited 16% more growth in their final 10 years than paired healthy trees (Thies, 1982). In the case of hollow stumps in the current study, we observed that growth of the pathogen appeared limited, survival of the tree did not appear threatened, and inoculum for passing the disease to the next stand did not seem to exist. We do not have additional information to explain our observations but speculate that it is due to resistance mechanisms within individual trees. This may be a useful characteristic for geneticists to use to develop trees resistant to LRR.

Observations on this and related studies lead us to suggest a need to examine more closely the epidemiology of LRR. The premise that *P. weirii* spreads only by root-to-root contact between host trees is supported by field observations and is appealing in that it explains the observed clusters or centers of infected trees that seem to expand at slow but sure rates. Some anomalies reported previously and some that appeared in this study suggest that the infection process is more complex and needs to be better understood. Anomalies from our five study sites included the following: (1) Scattered infected trees are difficult to explain based just on a model of an expanding infection center. (2) On our 239 plots, LRR sometimes appeared where it had not been before. By the conclusion of our study, of 47 plots without LRR (25 nonstumped plots and 22 stumped plots), nine plots in the replacement stand (six and three plots, respectively) had LRR-killed trees. In a study of individual trees, Thies and Nelson (1997) reported that of 45 ‘‘probably not infected’’ Douglas-fir (greater than 50 cm DBH) located more than 25 m from identifiable inoculum sources, 13 had LRR when their roots were excavated and thoroughly examined. Even with inoculum removal as complete as is reasonably possible, LRR still reappears on some sites (Morrison et al., 1988). (3) On some plots where inoculum was present in the parent stands the disease has not reappeared in the replacement stand. Of 94 non-

stumped plots with LRR in the parent stand, 29 did not have LRR-killed trees in the replacement stand. From an intensive examination of LRR distribution in a 60-year-old Douglas-fir stand and its 300-year-old parent stand, Tkacz and Hansen (1982) report instances of presumed healthy trees growing in previous infection centers. (4) Some infected stumps (hollow) do not seem likely to spread the disease to a subsequent stand. Observations such as these confound our understanding of disease spread and impact. If anomalies such as these are supported by additional field observations and other means of spread can be verified, then recommended management actions in stands with LRR may need revision.

5. Conclusions

In the current study we found:

1) Stumping consistently reduced Douglas-fir mortality from LRR in the replacement stand.
2) Using N fertilizer did not reduce Douglas-fir mortality from LRR in the succeeding stand.
3) Preplant N fertilization increased growth of Douglas-fir seedlings but only in some stands.
4) N fertilizer can have a long, positive residual effect on Douglas-fir growth.
5) Effects on soil of heavy equipment used in the stumping operations did not reduce subsequent Douglas-fir survival or growth.
6) There was 24.8% less wood volume in areas where LRR was present compared to areas where it was not.
7) Stumping plots with LRR increased volume 25.4%.
8) Both INOC, as used here, and a count of *P.weirii*-infected stumps in a parent stand, may have little value in predicting LRR-caused mortality of Douglas-fir in a replacement stand.

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